Approximation of areas

1 Use the trapezoidal rule to find an approximation for each integral:

a $\int_2^5 3x^2\, dx$ b $\int_0^2 (2x^3 - 1)\, dx$ c $\int_2^4 dx$

2 Find an approximation to $\int_0^1 x^3\, dx$ using the trapezoidal rule with:

a 1 subinterval b 2 subintervals

3 Given the table of values, find the approximate value of each definite integral:

a $\int_0^8 f(t)\, dt$

t	0	2	4	6	8
$f(t)$	3	7	11	9	3

b $\int_0^{40} f(x)\, dx$

x	0	10	20	30	40
$f(x)$	350	410	435	450	460

c $\int_{-4}^2 f(x)\, dx$

x	-4	-3	-2	-1	0	1	2
$f(x)$	0	4	5	3	10	11	2

4 Use the trapezoidal rule with 2 subintervals to find an approximation for the following to three decimal places:

a $\int_3^4 \ln 2x\, dx$ b $\int_0^6 \frac{dx}{x+3}$

5 Find an approximation to the following integrals, rounding your answers to three decimal places:

a $\int_2^3 e^{2x}\, dx$ using 3 subintervals b $\int_0^2 \left(4x^2 - 16\right)\, dx$ using 4 subintervals

c $\int_0^1 \sqrt{2x}\, dx$ using 5 subintervals d $\int_1^3 \frac{dx}{x^3}$ using 4 subintervals

e $\int_2^5 \frac{2}{3x-4}\, dx$ using 6 subintervals

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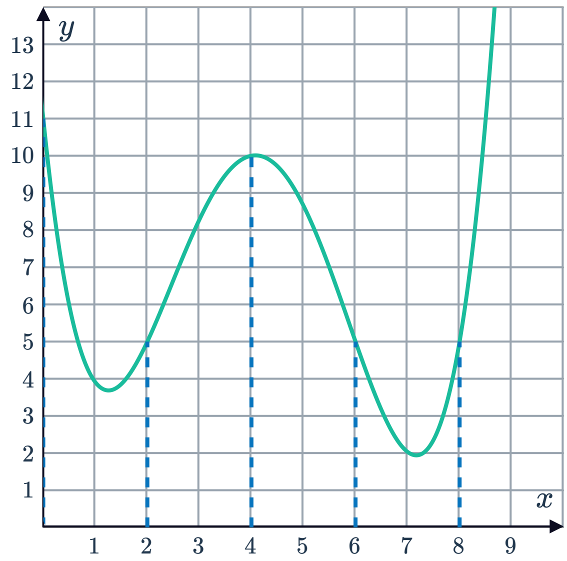
Use one application of the trapezoidal rul[redacted]pproximate $\int_2^3 \frac{e^x}{x}\,dx$ to one decimal place.

7 Use three applications of the trapezoidal rule to approximate $\int_0^9 e^{-x^2}\,dx$ to one decimal place.

8 Approximate $\int_0^8 8x\,dx$ by using four rectangles of equal width whose heights are the values of the function at the midpoint of each rectangle.

9 Approximate $\int_1^5 \frac{1}{x}\,dx$ by using four rectangles of equal width whose heights are the values of the function at the right endpoint of each rectangle.

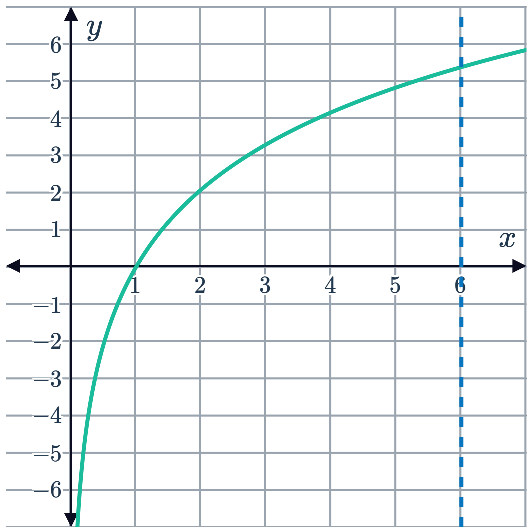
10 In the following graph, the interval $[0,8]$ is partitioned into four subintervals $[0,2]$, $[2,4]$, $[4,6]$, and $[6,8]$:



a Approximate the area A using rectangles for each subinterval whose heights are equal to the function values of the left side of the subintervals.

b Approximate the area A using rectangles for each subinterval whose heights are equal to the function values of the right side of the subintervals.

11 The function $y = 3\ln x$ has been graphed: Use two applications of the trapezoidal rule to approximate the area bound by the curve, the x-axis and and $x = 6$. Round your answer to one decimal place.



12 Consider the function $y = e^{x^2}$.

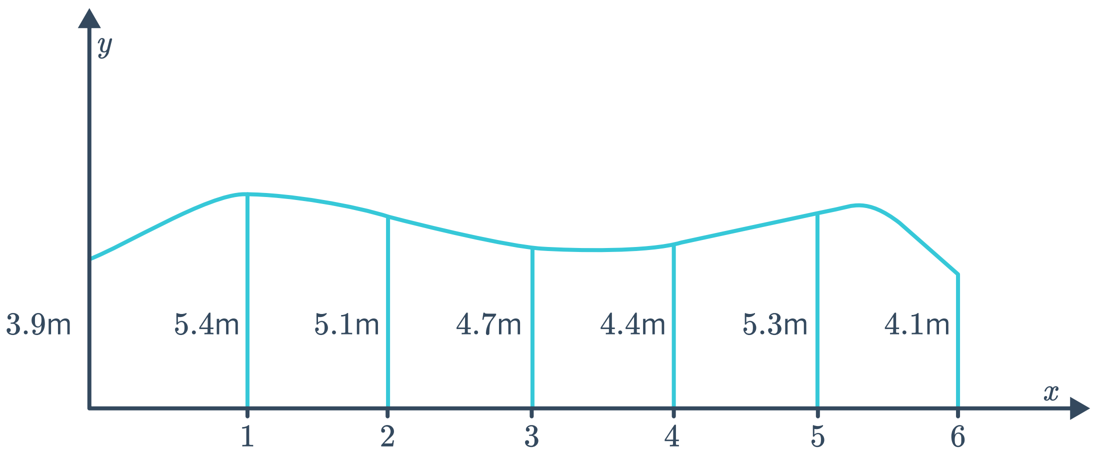
a Determine y''.

b Using two applications of the trapezoidal rule, approximate the area bound by the curve and the x-axis between $x = 0$ and $x = 2$. Round your answer to one decimal place.

c Is the approximation given by the trapezoidal rule an underestimate or an overestimate of the actual area? Explain your answer.

13 Consider the integral $\int_1^5 (\ln x + 4)\ dx$.
Approximate the area bounded by the curve, the x-axis, $x = 1$ and $x = 5$ using two applications of the trapezoidal rule. Round your answer to two decimal places.

14 Use the trapezoidal rule to find the approximate area of the irregular figure below:



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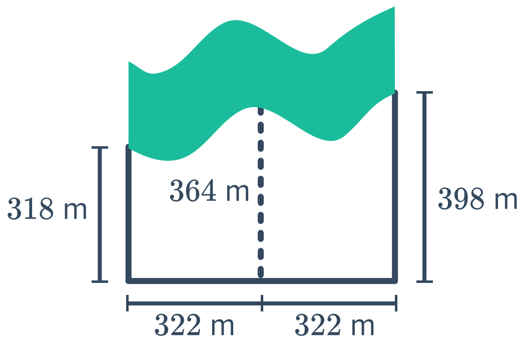
Applications

15 The following piece of land has straight boundaries on the east, west and south borders and is bounded by a creek to the north. The land has been divided into two sections so we can use the trapezoidal rule to approximate the area:

a. Find the approximate area of the piece of land by using two applications of the trapezoidal rule.

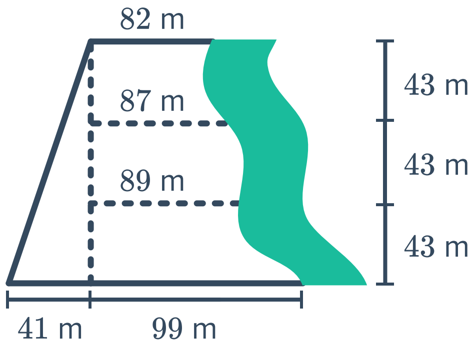
b. During a heavy storm, 35.2 mm of rain fell. Find the volume of water that falls on this land. Round your answer to the nearest cubic metre.



16 A surveyor made the following diagram with measurements for a property she was mapping out. On the west side of the property is a river:

a. Find the approximate total area of the property by using three applications of the trapezoidal rule.

b. The average weekly rainfall is 34 mm. Calculate the total volume of water that falls on the land in cubic metres. Round your answer to two decimal places.



17 A river has its depths marked out at equal intervals of 9 m. The depths are $0, 12, 14, 17, 5,$ and 0 m respectively. Find the approximate area of the cross section of the river.

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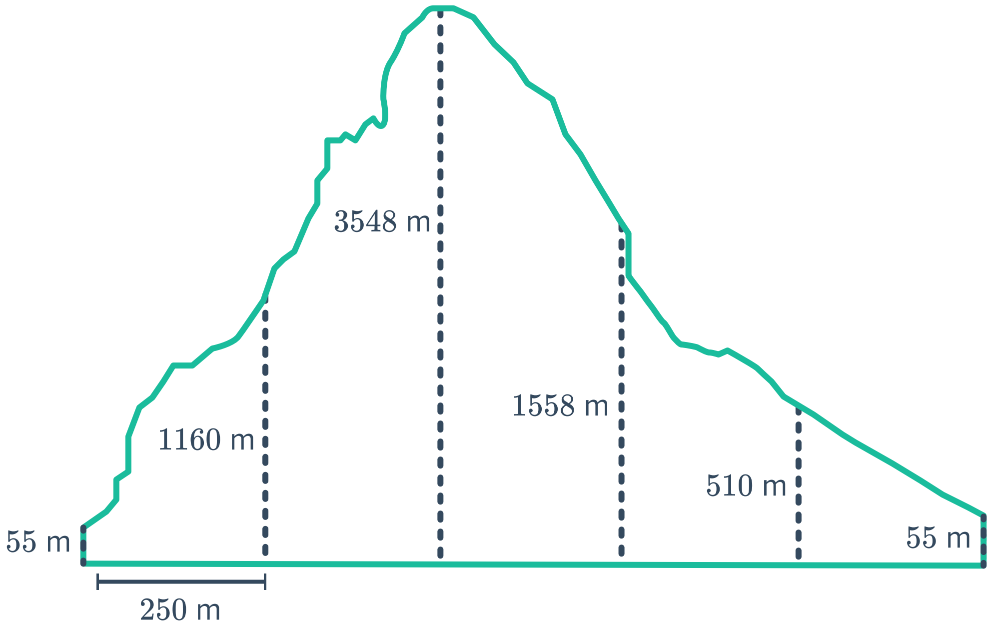
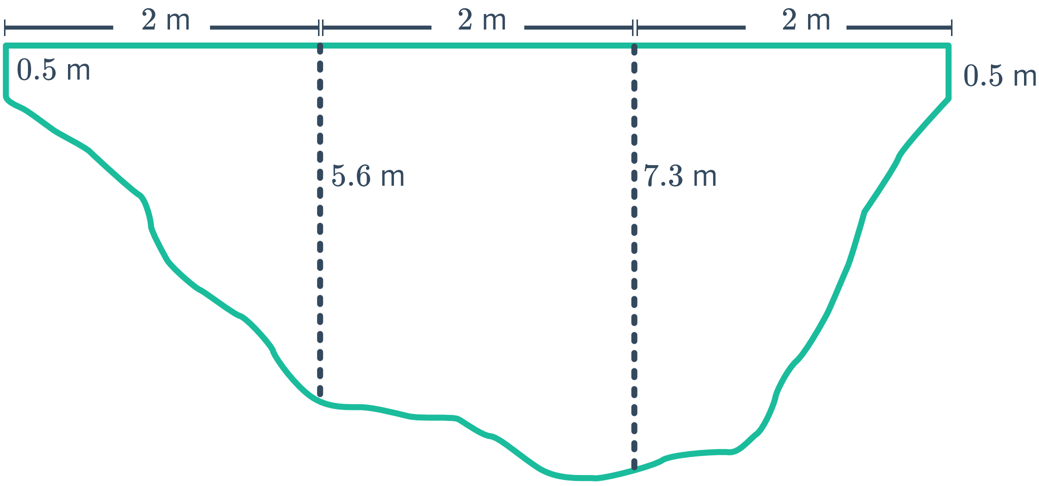
18 The elevation values of a mountain are re[redacted]d at equal intervals of 250 m. The heights are shown in the diagram:



Find the approximate area of the cross section of the mountain.

19 The diagram shows the cross-section of a river. The depths of the river are marked at 2-metre intervals:



Using three applications of the trapezoidal rule, approximate the area of the cross-section of the river to one decimal place.

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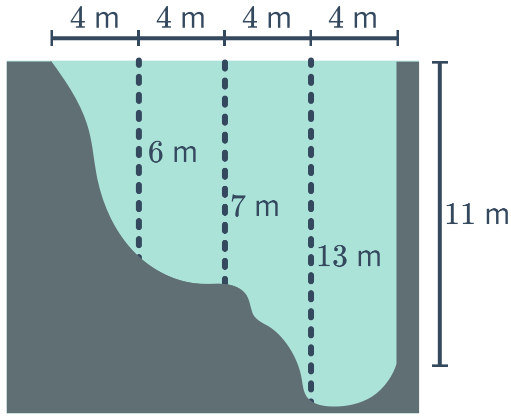
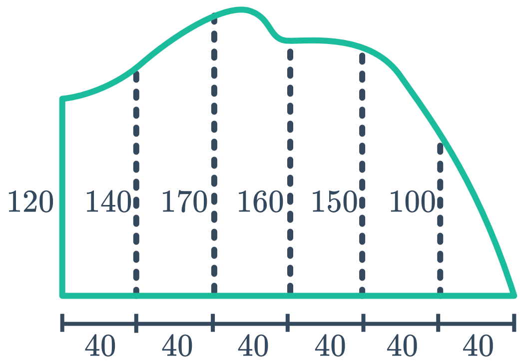
20 Use four applications of the trapezoidal rule to approximate the area of the cross-section of the following river:



21 The following shape has measurements given in metres. Use the trapezoidal rule to find the area in hectares.



22 A garden is 49 m long. At 7 m intervals, the width of the garden was given by the following measurements:

$$0 \text{ m}, 2.9 \text{ m}, 5.2 \text{ m}, 6.6 \text{ m}, 5.6 \text{ m}, 4.3 \text{ m}, 3 \text{ m}, 2.5 \text{ m}$$

Using the trapezoidal rule, approximate the area of the garden to two decimal places.

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